Z1_CONTROL

J5
Raspberry_Pi_4
GPIO12/PWM0
GPIO13/PWM1
GPIO14/UART_TXD
GPIO15/UART_RXD
GPIO16/SPI1_CE2
GPIO17/SPI1_CE1
GPIO18/SPI1_CE0/PCM_CLK/PWM0
GPIO19/SPI1_MISO/PCM_FS
GPIO20/SPI1_MOSI/PCM_DIN/PWM1
GPIO21/SPI1_SCLK/PCM_DOUT
GPIO22/SDIO_CLK
GPIO23/SDIO_CMD
GPIO24/SDIO_DAT0
GPIO25/SDIO_DAT1
GPIO26/SDIO_DAT2
GPIO27/SDIO_DAT3
5V
3V3
ID_SD_I2C0/GPIO00
ID_SC_I2C0/GPIO01
SDA_I2C1/GPIO02
SCL_I2C1/GPIO03
GPCLK0/GPIO04
GPCLK1/GPIO05
GPCLK2/GPIO06
CE1_SPI0/GPIO07
CE0_SPI0/GPIO08
MISO_SPI0/GPIO09
MOSI_SPI0/GPIO10
SCLK_SPI0/GPIO11
GND
+5V_Z1
Z1_DEG_A
Z1_DEG_B
Z1_CAN_INT_A
Z1_CAN_INT_B
Z1_ERR
UDP_RX
Z1_CMD_TX
CMD_TX
UDP_RX
Z1_CAN_TX_A
Z1_CAN_TX_B
CAN_TX_A
CAN_TX_B
Z1_CS_CAN_B
Z1_CS_CAN_A
Z1_MISO
Z1_MOSI
Z1_SCLK
TP1
TP2
TP3
TP4
D19
DEGRADED
470
R35
D20
DEGRADED
470
R36
D23
ERROR
470
R39

Z2_CONTROL

Notes:
– AGND and GND are shared
– This layout doesn't have pin 31 or 47, but they should be tied to ground and that seems correct on the PCB. They are potentially supposed to have a 2.2 uF cap
– The BMkz oscillator is not connected
S3
RESET
C72
1uF
C70
0.1uF
Z3_NRST
Steering (Zone 2)
STM_STMPWR
U7
STM32F401RETx
NRST
BOOT0
VREF+
PH0
PH1
PD2
PC0
PC1
PC2
PC3
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PC4
PC5
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
PC6
PC7
PC8
PC9
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PC10
PC11
PC12
PC13
PC14
PC15
VCAP1
VSS
VDD
VBAT
SERVO_PWM
Z2_DEG_A
Z2_USART_TX
Z2_USART_RX
Z2_DEG_B
FR_BLINK
Z2_CAN_TX_A
Z2_INT_CAN_B
Z2_CAN_RX_B
Z2_CAN_RX_A
Z2_TMS
Z2_TCK
Z2_ERR
Z2_SCLK
Z2_MISO
Z2_MOSI
Z2_CS_CAN_A
Z2_CAN_TX_B
FL_BLINK
Z2_INT_CAN_A
Z2_CAN_TX_A1
CMD_TX
Z2_CAN_RX_B
Z2_CAN_RX_A
TP7
TP6
TP8
TP32
Z3_ERR
Z2_SWO
TP33
Z2_CAN_RX_B
Z2_CAN_TX_B
Z2_CS_CAN_A
Z2_CS_CAN_B
Z2_MISO
Z2_MOSI
Z2_SCLK
Z2_TEST_BUTTON
Z2_RELAY_TX
SERVO_CURRENT
Z3_BOOT0
Z3_MCO
TP29
TP30
TP9
R66
10k
R67
0
R68
0
Y5
8MHz
C68
4.3pF
C69
4.3pF
C71
4.7uF
BEAD
FB2
J3
Z2_SWD
J6
Z2_USART
Z2_TCK
Z2_TMS
Z2_NRST
Z2_SWO
Z2_USART_TX
Z2_USART_RX
LS1
BUZZER
S4
TEST
R69
4K7
R70
120
C73
0.1uF
Z2_TEST_BUTTON
D9
DEGRADED
470
R25
D10
DEGRADED
470
R26
D11
FL_BLINKER
470
R27
D12
FR_BLINKER
470
R28
D13
ERROR
470
R29
Z2_DEG_A
Z2_DEG_B
FL_BLINK
FR_BLINK
Z2_ERR

Z3_CONTROL

Notes:
– AGND and GND are shared
– This layout doesn't have pin 31 or 47, but they should be tied to ground and that seems correct on the PCB. They are potentially supposed to have a 2.2 uF cap
– The BMkz oscillator is not connected
S2
RESET
C62
1uF
C64
0.1uF
Z3_NRST
STM_STMPWR
U3
STM32F401RETx
NRST
BOOT0
VREF+
PH0
PH1
PD2
PC0
PC1
PC2
PC3
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PC4
PC5
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
PC6
PC7
PC8
PC9
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PC10
PC11
PC12
PC13
PC14
PC15
VCAP1
VSS
VDD
VBAT
ENC_LEFT_B
ENC_RIGHT_A
Z3_USART_TX
Z3_USART_RX
ENC_RIGHT_B
RL_BLINK
MOTOR_LEFT_PWM
Z3_INT_CAN_B
ENC_LEFT_A
MOTOR_RIGHT_PWM
MOTOR_PWM_OUT
TP15
MOTOR_LEFT_DIR
PWM_SET
MOTOR_DIR
MOTOR_CURRENT_LEFT
MOTOR_CURRENT_RIGHT
Z3_CAN_TX_B
Z3_CAN_TX_A
Z3_CS_CAN_B
Z3_CAN_RX_B
Z3_CAN_RX_A
Z3_DEG_A
Z3_DEG_B
Z3_ERR
Z3_TEST_BUTTON
MOTOR_RIGHT_DIR
Z3_SCLK
Z3_MISO
Z3_MOSI
Z3_CS_CAN_A
DIR_A
RR_BLINK
Z3_INT_CAN_A
Z3_SWO
TP27
Z3_BOOT0
Z3_MCO
TP28
TP14
TP13
TP17
Z3_CAN_TX_B
TP16
Z3_CAN_TX_A
TP19
Z3_CAN_RX_B
TP18
Z3_CAN_RX_A
TP31
Z3_ERR
R65
10k
R64
4K7
R63
120
C63
0.1uF
R8
0
R9
0
Y7
8MHz
C65
4.3pF
C66
4.3pF
C67
4.7uF
BEAD
FB1
J2
Z3_SWD
J4
Z3_USART
Z3_TCK
Z3_TMS
Z3_NRST
Z3_SWO
Z3_USART_TX
Z3_USART_RX
LS2
BUZZER
S1
TEST
D14
DEGRADED
470
R30
D15
DEGRADED
470
R31
D16
RL_BLINKER
470
R32
D17
RR_BLINKER
470
R33
D18
ERROR
470
R34
Z3_DEG_A
Z3_DEG_B
RL_BLINK
RR_BLINK
Z3_ERR

U4
ACS712xLCTR-05B
VCC
VIOUT
FILTER
IP+
IP-
GND
C76
1000pF
C77
0.1uF
MOTOR_CURRENT_RIGHT
TP46
L_RIGHT
U2
ACS712xLCTR-05B
MOTOR_CURRENT_LEFT
TP45
L_LEFT
+12V
+5V_Z3
C26
470 uF
C25
470 uF
U20
L298N
IN1
IN2
ENA
IN3
IN4
ENB
OUT1
OUT2
OUT3
OUT4
SENSE_A
SENSE_B
MOTOR_RIGHT_PWM
MOTOR_RIGHT_DIR
MOTOR_LEFT_PWM
MOTOR_LEFT_DIR
J7
MOTOR_RIGHT
ENC_RIGHT_A
ENC_LEFT_B
J8
MOTOR_LEFT
ENC_LEFT_A
ENC_LEFT_B

Z1_CANB

U13
MCP2515-xST
U14
SN65HVD230
J11
CANA_Z1
Y4
CX3225SB16000H0FLJCC
C13
22pF
C14
22pF
R12
10k
R13
47K
C15
100nF
C16
100nF
Z1_CANH_B
Z1_CANL_B
TP35
TP36
Z1_MOSI
Z1_MISO
Z1_CS_CAN_B
Z1_SCLK
Z1_INT_CAN_B
CAN controller schematic copied from: https://wiki.seeedstudio.com/xiao-can-bus-expansion

Z2_CANB

U15
MCP2515-xST
U16
SN65HVD230
J12
CANA_Z1
Y5
CX3225SB16000H0FLJCC
C17
22pF
C18
22pF
R14
10k
R15
47K
C19
100nF
C20
100nF
Z2_CANH_B
Z2_CANL_B
TP37
TP38
Z2_MOSI
Z2_MISO
Z2_CS_CAN_B
Z2_SCLK
Z2_INT_CAN_B
CAN controller schematic copied from: https://wiki.seeedstudio.com/xiao-can-bus-expansion

Z3_CANB

U17
MCP2515-xST
U18
SN65HVD230
J14
CANA_Z1
Y6
CX3225SB16000H0FLJCC
C21
22pF
C22
22pF
R16
10k
R17
47K
C23
100nF
C24
100nF
Z3_CANH_B
Z3_CANL_B
TP41
TP42
Z3_MOSI
Z3_MISO
Z3_CS_CAN_B
Z3_SCLK
Z3_INT_CAN_B
CAN controller schematic copied from: https://wiki.seeedstudio.com/xiao-can-bus-expansion

JP1
JP9
R18
120
R19
120
CANH_A
CANH_A
JP11
JP19
R20
120
R21
120
CANH_B
CANH_B

U5
INA181
C78
0.1uF
R71
10hm
J9
SERVO
SERVO_CURRENT
SERVO_PWM
TP47
SERVO_CURRENT

Z1_CANA

U1
MCP2515-xST
U8
SN65HVD230
J10
CANA_Z1
Y1
CX3225SB16000H0FLJCC
C2
22pF
C3
22pF
R1
10k
R2
47K
C4
100nF
C1
100nF
Z1_CANH_A
Z1_CANL_A
TP34
TP33
Z1_MOSI
Z1_MISO
Z1_CS_CAN_A
Z1_SCLK
Z1_INT_CAN_A
CAN controller schematic copied from: https://wiki.seeedstudio.com/xiao-can-bus-expansion

Z2_CANA

U9
MCP2515-xST
U10
SN65HVD230
J13
CANA_Z1
Y2
CX3225SB16000H0FLJCC
C5
22pF
C6
22pF
R3
10k
R4
47K
C7
100nF
C8
100nF
Z2_CANH_A
Z2_CANL_A
TP39
TP40
Z2_MOSI
Z2_MISO
Z2_CS_CAN_A
Z2_SCLK
Z2_INT_CAN_A
CAN controller schematic copied from: https://wiki.seeedstudio.com/xiao-can-bus-expansion

Z3_CANA

U11
MCP2515-xST
U12
SN65HVD230
J15
CANA_Z1
Y3
CX3225SB16000H0FLJCC
C9
22pF
C10
22pF
R5
10k
R6
47K
C11
100nF
C12
100nF
Z3_CANH_A
Z3_CANL_A
TP43
TP44
Z3_MOSI
Z3_MISO
Z3_CS_CAN_A
Z3_SCLK
Z3_INT_CAN_A
CAN controller schematic copied from: https://wiki.seeedstudio.com/xiao-can-bus-expansion

J1
XT60-M
F1
15 A
Q2
IRF4905
D1
15V
R7
10k
+12V
TP10
12V_IN

+5V_Z1
D2
WS2812B
D3
WS2812B
D4
WS2812B
D5
WS2812B
D6
WS2812B
D7
WS2812B
D8
WS2812B
D21
WS2812B
D22
WS2812B
D24
WS2812B
D25
WS2812B
D26
WS2812B
D27
WS2812B

Z1_3_3_REG

IC3
TPS564252DRLR
C43
330pF
C44
330pF
R45
510K
R46
68k
R48
45k
R50
53
R49
10k
R47
10k
L3
2.2uH
C41
22uF
C42
0.1uF
C45
22uF
C46
22uF
C47
0.1uF
TP21
Z1_3_3_EN
TP22
Z1_3_3_OUT
JP25

Z2_3_3_REG

IC1
TPS564252DRLR
C33
330pF
C32
330pF
R23
510K
R24
68k
R22
45k
R37
53
R11
10k
R10
10k
L1
2.2uH
C27
22uF
C28
0.1uF
C29
22uF
C30
22uF
C31
22uF
C32
0.1uF
TP5
Z2_3_3_EN
TP11
Z3_3_3_OUT
JP21

Z3_3_3_REG

IC2
TPS564252DRLR
C36
330pF
C37
330pF
R38
510K
R40
68k
R42
45k
R44
53
R41
10k
L2
2.2uH
C34
22uF
C35
0.1uF
C38
22uF
C39
22uF
C40
0.1uF
TP12
Z3_3_3_EN
TP20
Z3_3_3_OUT
JP23

Z1_5_REG

IC4
TPS564252DRLR
C58
330pF
C57
330pF
R51
510K
R52
68k
R54
220k
R56
53
R55
30k
R53
10k
L4
4.7uH
C48
22uF
C49
0.1uF
C52
22uF
C53
22uF
C54
0.1uF
TP23
Z1_5_EN
TP24
Z1_5_OUT
JP27

Z2_7_REG

IC6
TPS564252DRLR
C81
10nF
C82
330pF
R72
510K
R73
68k
R75
330k
R77
53
R76
30k
R74
10k
L6
4.7uH
C79
22uF
C80
0.1uF
C83
22uF
C84
22uF
C85
0.1uF
TP48
Z3_5_EN
TP49
Z2_5_OUT
JP33

Z3_5_REG

IC5
TPS564252DRLR
C57
330pF
C58
330pF
R57
510K
R58
68k
R60
220k
R62
53
R61
30k
R59
10k
L5
4.7uH
C55
22uF
C56
0.1uF
C59
22uF
C60
22uF
C61
0.1uF
TP25
Z3_5_EN
TP26
Z3_5_OUT
JP29

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Rev: 1.0
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